Midterm 2 Study Sheet

Show reasoning, give exact answers, and include units in applications. Complete all work on separate paper, including sketches and sign charts. Parts within each problem are independent unless stated otherwise. The final problem offers additional challenge.

1. Compute the following limits. Give a finite value, $+\infty$, $-\infty$, or DNE. Identify any indeterminate form and show how you resolve it.

   (a) $\lim_{x\to 2}(3x^2 - x + 4)$

   (b) $\lim_{z\to 1}\dfrac{\sqrt{7+2z}}{1+2^z}$

   (c) $\lim_{x\to 3}\dfrac{x^2-9}{x-3}$

   (d) $\lim_{t\to 0}\dfrac{\sqrt{9+t}-3}{t}$

   (e) $\lim_{x\to+\infty}\left(\sqrt{x^2+8x}-x\right)$

   (f) $\lim_{x\to 2^+}\left(\dfrac{1}{x-2}-\dfrac{4}{x^2-4}\right)$

   (g) $\lim_{t\to 1^-}\dfrac{1}{t-1}$

   (h) $\lim_{x\to+\infty}\dfrac{5x^2-x+1}{2x^2+3}$

   (i) Let $J(x) = \dfrac{x^2-1}{x^2-x-2}$. Compute each limit below. Use your answers to identify the hole and all horizontal and vertical asymptotes of J.

   $$\lim_{x\to -1} J(x) \quad \lim_{x\to 2^-} J(x) \quad \lim_{x\to 2^+} J(x)$$
   $$\lim_{x\to 2} J(x) \quad \lim_{x\to -\infty} J(x) \quad \lim_{x\to +\infty} J(x)$$

2. Limits and continuity.

   (a) Let

   $$F(x) = \begin{cases} \dfrac{\sqrt{x+5}-3}{x-4}, & x \geq -5,\ x \neq 4, \\ k, & x = 4. \end{cases}$$

   Find $\lim_{x\to 4} F(x)$ and the value of k that makes F continuous at 4.

   (b) Find k so that the function below is continuous on $\mathbb{R}$. Justify your answer using one-sided limits.

   $$B(x) = \begin{cases} x+2, & x < 1, \\ 2x+k, & x \geq 1. \end{cases}$$

3. Find each derivative with respect to x. Work where the function is defined and differentiable. You may leave products and quotients unsimplified.

   (a) $2x^5 - 3\sqrt{x} + \dfrac{4}{x}$

   (b) $(2x-1)^5$

   (c) $(x^2+2)e^{-x}$

   (d) $\dfrac{x^2+1}{x+2}$

   (e) $\ln(3x^2+1)$

   (f) e^{x^2-2x}

   (g) $\sqrt{4x+1}$

   (h) $\dfrac{\ln x}{x}$

(i) $\sqrt{\ln x + x^2}$

(j) $e^{3x}(4x+5)^6$

(k) $7x^4 - 3x + 2$

(l) $\dfrac{1}{x}$

(m) $\sqrt{x}$

(n) $\ln x$

(o) e^x

(p) $(5x-2)^3$

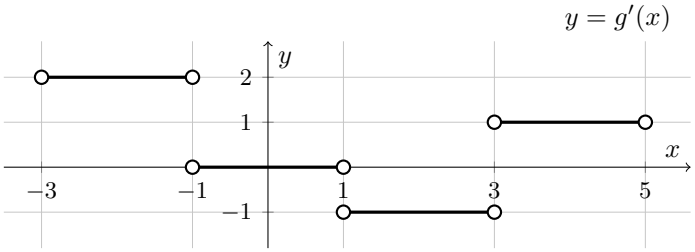
(q) $(2x+1)\ln x$

(r) $\dfrac{4x}{x^2+1}$

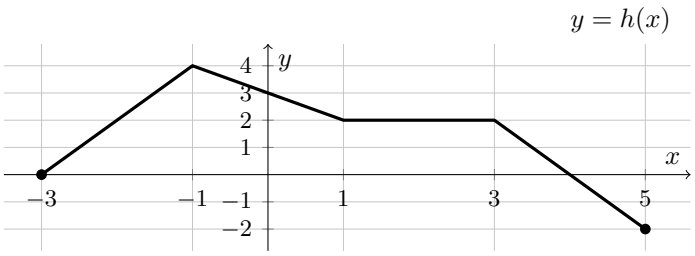
(s) $\ln\left(\sqrt{x^2+4}\right)$

(t) $\dfrac{e^{-2x}}{x+1}$

4. Find an equation of the tangent line to each graph at the given point.
   (a) $y = \sqrt{x+5}$ at $(4,3)$.
   (b) $y = e^{2x}$ at $(0,1)$.
   (c) $y = \ln(x+1)$ at $(e-1,1)$.
   (d) $y = \dfrac{x^2+1}{x+1}$ at $(1,1)$.
5. Linear approximation and marginal quantities.
   (a) A differentiable function satisfies $F(3) = 7$ and $F'(3) = -2$. Use a linear approximation to estimate $F(2.9)$.
   (b) A company's total cost is $C(q)$ dollars for q items. Given $C(150) = 4200$ and $C'(150) = 18$, estimate $C(154)$ and interpret $C'(150)$ with units.
   (c) A company sells q items at a price $p(q)$ dollars per item. Given $p(50) = 30$ and $p'(50) = -.20$, find the marginal revenue at $q = 50$.
6. Functions and their derivatives.
   (a) A continuous function g on $[-3,5]$ has $g(-3) = -1$. Its derivative is shown below wherever it exists. Sketch g.



   (b) The graph of h consists of straight line segments. Sketch h' on $(-3,5)$ and indicate where h' does not exist.



7. Critical numbers and classification.

   (a) Find all critical numbers of $u(x) = x^2 e^{-x}$ on $\mathbb{R}$. You do not need to classify them.

   (b) Find and classify the critical numbers of $v(x) = \ln x - x/2$ on its domain.

   (c) Find all critical numbers of $p(x) = x^3 + 3x^2 - 9x + 1$ and classify each.

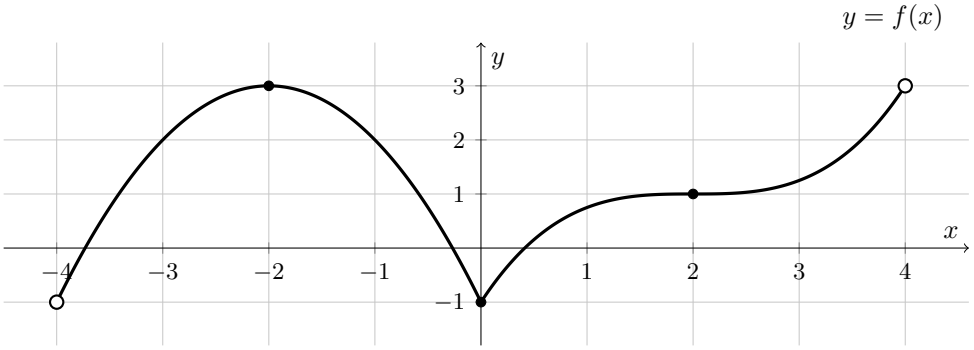
   (d) A differentiable function G on $\mathbb{R}$ satisfies $G'(x) = (x+2)(x-1)^2(x-3)$. Find and classify all critical numbers of G. Label all critical numbers of G on the number line, and complete the sign chart for G'.

$G'(x)$

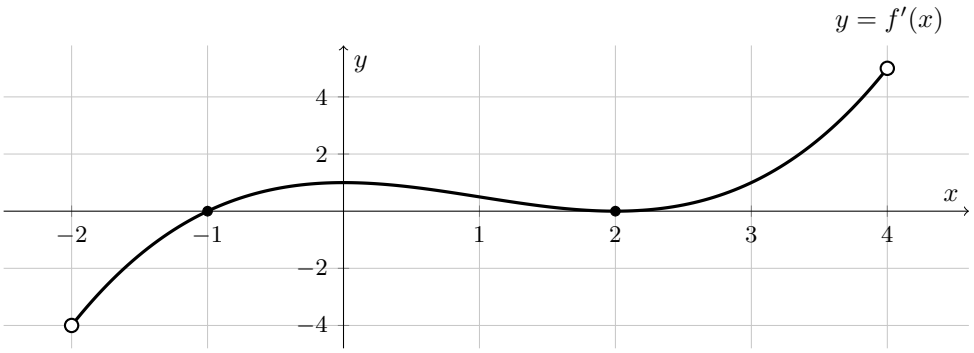
x ⟷

8. Graphical classification of critical numbers.

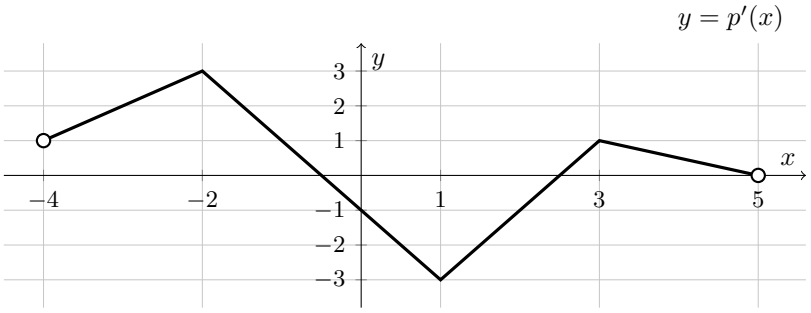
   (a) List and classify all critical numbers of the function shown, considering only $-4 < x < 4$.



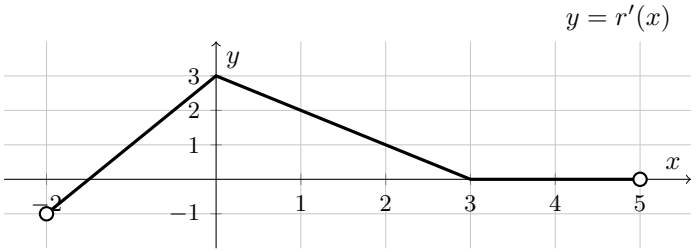
   (b) The graph shows f' for a differentiable function f. Find and classify all critical numbers of f in $(-2, 4)$ using the signs of f'.



9. The graph shows p' for a differentiable function p on $(-4, 5)$. Determine where p is concave up or down, and give the x-coordinates of its inflection points. The segments are straight lines.



10. The graph of r' consists of straight line segments. Sketch r'' on $(-2, 5)$ and indicate where r'' does not exist.



11. Determine the concavity intervals and all inflection points of $w(x) = x^4 - 6x^2$.

12. Sketch $f(x) = x^3 - 6x^2 + 9x + 1$. Label its local extrema and inflection point, and show its increasing/decreasing behavior and concavity.

13. Additional challenge. Sketch $H(x) = \dfrac{\ln x}{x}$ on its domain. Label its local extrema, inflection points, and asymptotes, and show its monotonicity and concavity.

Derivative reference

$f'(a)$ is the instantaneous rate of change of f and the slope of its tangent line at $(a, f(a))$. Here c is a constant; rules apply where the functions are defined and differentiable.

$$\frac{d}{dx}x^n = nx^{n-1} \quad \frac{d}{dx}e^x = e^x \qquad \frac{d}{dx}\ln x = \frac{1}{x}\ (x > 0) \quad \frac{d}{dx}c = 0$$

Rewrite roots and reciprocals as powers: $\sqrt{x} = x^{1/2}$ and $1/x^m = x^{-m}$.

$$(cf \pm g)' = cf' \pm g' \qquad (fg)' = f'g + fg'$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2} \quad (g \neq 0) \quad \frac{d}{dx}f(g(x)) = f'(g(x))\,g'(x)$$

Tangent line at $x = a$: $y - f(a) = f'(a)(x - a)$.

Linear approximation: $f(x) \approx f(a) + f'(a)(x - a)$ (x near a).